Exploration of Highly Parallelizable Optimization Algorithms for Control

Semester Project

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Chapter 1

Introduction

Many recent problems of interest are solved by means of optimization algorithms that are able to manage a multitude of agents or handle efficiently large data sets. Among the problems of interest, many can be cast as convex program with a distributed structure. First, large-scale problems can be solved using appropriate parallel/decomposition-based optimization algorithms in order to decrease the solving time. Second, some applications require that some information remain private while solving an optimization problem. Solving the problem in a distributed or decentralized fashion helps to satisfy this particular constraint. Third, by making local copies of the data, information can be preserved in many locations, making the algorithm consequently more robust towards eventual failures.

1.1 Splitting method for distributed optimization

The objective of the present work is to solve an optimization problem that is distributed between various agents and a central operator. Each agent has an individual or selfish objective, while the central operator prompts them to cooperate to achieve a common objective. The problem must be solved in a decentralized and parallel fashion, while limited information between the central operator and the agents should be communicated. To solve the problem, a particular splitting method is used to distribute parts of the global problem over the agents while their privacy is being protected.

The Generalized Forward-Backward Splitting (GFBS) algorithm presented by H. Raguet and L. Landrieu in [8] is considered. The objective is to understand the algorithm and show why it is well suited for distributed optimization problems with limited information sharing. From the theory of proximal operator[7], the algorithm can be compared to the proximal gradient method. This splitting method is discussed and illustrated as a fixed-point method. Finally, the GFBS algorithm is presented and compared to another splitting method.

1.2 Micro-grid case study

The interest for micro-grids has been growing fast for the recent years due the wide-spread use of renewable energies. Micro-grids interconnect loads with energy generation and storage systems. The energy production and storage sites being closer to the demand allow for better energy management improving consequently its quality and minimizing the losses related to transport. Micro-grids can operate completely separated from the grid in so called islanded applications. Connected to the central grid, they can be used for frequency stabilisation applications[1].

A micro-grid can be modelled as the interconnection of loads, energy production systems and energy storage devices. In our case study, the loads are represented by offices buildings that are able to adapt their energy consumption depending on the available power. Regarding the energy storage devices, the case study is limited to batteries and flywheels and their advantages and drawbacks are discussed. Regarding the energy generation part, there are several ways of producing energy locally using renewable or non-renewable energy resources. The case-study is limited to wind power and solar power. For the purpose of the simulation, the micro-grid has to manage all of its agents to track a given power profile. All of the components are then first modelled and integrated to problem. Then, the GFBS algorithm solves the distributed optimization problem for different power profiles.

Chapter 2

Splitting method

In the present chapter, the Generalized Forward-Backward Splitting algorithm is presented. In order to better understand the working principle of the algorithm, several points are discussed. First, the GFBS algorithm is a proximal splitting method and it is therefore useful to recall the definition of the proximal operator. Second, a particular case of the GFBS algorithm can be compared to the proximal gradient method, which is also a forward-backward method and has an interesting fixed-point method interpretation. Third, the algorithm of the GFBS method is presented into details together with the assumption done one the specific optimization problem it solves. Finally the algorithm is compared to another splitting method, namely the Alternating Direction Method of Multipliers (ADMM).

2.1 Problem statement

The optimization problem considered in the present work is an operator-agent optimization problem with limited information sharing between the agents and the operator. In this section, the formulation of the problem is presented as well as the issue related to the agents' privacy.

2.1.1 Problem formulation

Consider the following problem

$$
\begin{aligned}
\underset{y,u,x}{\text{minimize}} \quad & \sum_{t=1}^{T} \frac{\sigma}{2}\Big(\sum_{i=1}^{M} y_i(t) - r(t)\Big)^2 + \sum_{t=1}^{T}\sum_{i=1}^{M} q(t) y_i(t) \\
\text{subject to} \quad & x_i(t+1) = A_i x_i(t) + B_i u_i(t) \\
& y_i(t) = C_i x_i(t) + D_i u_i(t) \\
& x_i(t) \in X_i \\
& u_i(t) \in U_i \\
& i = 1, \cdots, M \\
& t = 1, \cdots, T
\end{aligned}
\quad , \tag{2.1}
$$

where $y_i(t), q(t), r(t) \in \mathbb{R}$, $y_i \in \mathbb{R}^T$ and $y \in \mathbb{R}^{TM}$. The problem is solved for M agents over a time horizon T. Each agent is represented by a linear discrete-time model with state $x_i(t) \in \mathbb{R}^{p_i}$ and input $u_i \in \mathbb{R}^{m_i}$. They must satisfy the dynamic imposed by the state matrix $A_i \in \mathbb{R}^{p_i \times p_i}$ and the input matrix $B_i \in \mathbb{R}^{p_i \times m_i}$ of the model. The throughput matrix for each agent is $D \in \mathbb{R}^{p_i}$ and the output matrix is $D \in \mathbb{R}^{m_i}$. In addition, the state and the input are respectively constrained to the set $X_i \in \mathbb{R}^{p_i}$ and $U_i \in \mathbb{R}^{m_i}$ over the entire time horizon.

The cost function for the minimization problem states that the aggregated output of the agents has to match a given target, denoted by the parameter r, while they minimize their personal goal over the entire horizon. In other words, a central operator must bring the agents to collaborate in order to achieve a common goal r while the agents maximize their personal benefit. The parameter σ is adjusted by the operator to increase or decrease the collaboration between the agents while the cost of achieving the personal goal is influenced by parameter q. Note that q can be adjusted over the time horizon.

The problem (2.1) can be re-formulated in a more compact way

$$
\begin{aligned}
\underset{y,u,x}{\text{minimize}} \quad & \frac{1}{2}\left(h^\top y - r\right)^\top H \left(h^\top y - r\right) + \sum_{i=1}^{M} q^\top y_i \\
\text{subject to} \quad & x_i(t+1) = A_i x_i(t) + B_i u_i(t) \\
& y_i(t) = C_i x_i(t) + D_i u_i(t) \\
& x_i(t) \in X_i \\
& u_i(t) \in U_i \\
& i = 1, \cdots, M \\
& t = 1, \cdots, T
\end{aligned}
\quad , \tag{2.2}
$$

where $q \in \mathbb{R}^T, r \in \mathbb{R}^T, h \in \mathbb{R}^{TM \times T}$ and $H \in \mathcal{S}^T_{++}$. More specifically, the matrices h and H can be expressed according to the identity matrix $I \in \mathbb{R}^{T \times T}$

$$
H = \sigma I, \quad h = \left[I_1 \cdots I_M\right]^\top . \tag{2.3}
$$

2.1.2 Privacy

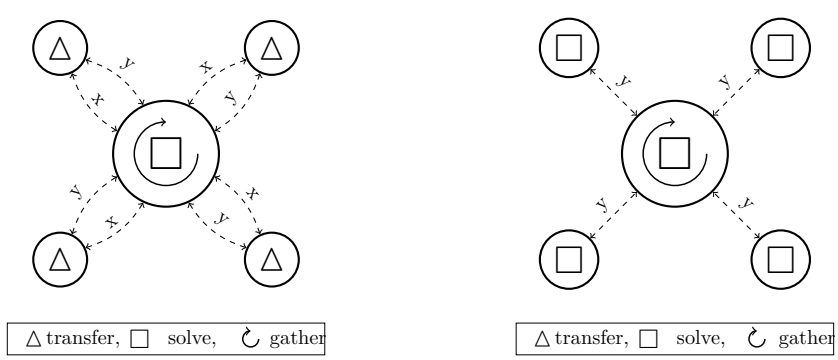
The agents have private information they don't want to communicate to the central operator, namely the model data such as the state and input matrices but also the state and input variables. In order to respect the privacy of each agent, an appropriate way for solving problem (2.2) must be found. Consider for example the following generic optimization problem

$$\underset{x,y}{\text{minimize}} \quad f(y) + \sum_i g_i(x, y) \ , \tag{2.4}$$

where y is a public variable and x a private variable related to the agents that should not be shared to the central operator. Problem (2.4) is similar to (2.2). The operator must manage the agents in such a way that the common goal denoted by f is achieved while every agent addresses its personal goal denoted by g_i. However, limited information should be provided to the operator and a clear distinction must be done between the private and public variables.

The privacy problem is illustrated by figure 2.1. For a centralized solving approach, every agent provides its variables x and y to the central operator. The optimization problem is then solved in a centralized fashion as illustrated in figure 2.1a. For a distributed solving approach, the variable x is not communicated to the central operator while y is well communicated. Then, each agent has to solve locally an optimization problem as illustrated in figure 2.1b.

In other words, to ensure the privacy of each agent, the optimization problem is split and distributed over the operator and all the agents. Besides the issue of privacy, splitting the problem can be beneficial for two reasons. First, instead of solving a complex central optimization problem, it is split into several sub-problems potentially easier to solve in terms of computational time. Second, the communication between the central operator and all the agents is facilitated because less variables must be communicated in comparison to the centralized solving approach. This represents a practical advantage in the case of a wire-less communication where bandwidth is consequently spared.



(a) Centralized solving approach (b) Distributed solving approach

Figure 2.1: Centralized vs. distributed solving approach

2.2 Proximal operator

The proximal operator is an efficient tool to solve large-scale non-smooth convex optimization problems and plays a central role in the present project. This section aims therefore at recalling the definition of the proximal operator. First, its definition is given and discussed. Second, an interpretation of the proximal operator as the resolvent of the subdifferential operator is explained.

2.2.1 Definition

The proximal operator $\mathbf{prox}_f : \mathbb{R}^n \to \mathbb{R}^n$ is defined by

$$\mathbf{prox}_f(v) = \underset{x}{\operatorname{argmin}}\left(f\left(x\right) + \frac{1}{2}\|x - v\|_2^2 \right), \tag{2.5}$$

where $f : \mathbb{R}^n \to \mathbb{R} \cup \{+\infty\}$ should be convex, closed and proper. Due to the strong convexity of the quadratic term, the proximal operator has a unique minimizer for every $v \in \mathbb{R}^n$. The result of a proximal operator represents the best compromise of minimizing f and being near to v[7]. In order to control or to scale the trade-off between f and the quadratic term, a parameter λ strictly positive is defined such that the proximal operator becomes

$$\mathbf{prox}_{\lambda f}(v) = \underset{x}{\operatorname{argmin}}\left(f\left(x\right) + \frac{1}{2\lambda}\|x - v\|_2^2 \right). \tag{2.6}$$

A low value of λ suggest that the proximal operator yields a result close to v while a high value of λ suggests a result close to the minimum of f because it penalizes the quadratic term of the operator. Therefore, the parameter λ can be interpreted as a weight factor that influences the decision-making process of the proximal operator. In order to preserve convexity λ should obviously never become negative.

2.2.2 Resolvent of a subdifferential operator

The proximal operator can be interpreted as being the resolvent of a subdifferential operator[7]. The resolvent R of a relation or operator A on $\mathbb{R}^n$ is defined as[9]

$$R = (I + \alpha A)^{-1}, \tag{2.7}$$

where the scalar coefficient $\alpha \in \mathbb{R}$. Therefore, the proximal operator as resolvent of the subdifferential operator yields

$$\mathbf{prox}_{\lambda f} = \left(I + \lambda \partial f\right)^{-1}. \tag{2.8}$$

This can be shown easily from the definition of a resolvent and following the method used in[9]. Assume that f is closed convex and proper, α strictly positive and assuming $z, v \in \mathbb{R}^n$ such that z and v satisfy

$$z = (I + \alpha \partial f)^{-1}(v). \tag{2.9}$$

Statement (2.9) can be reformulated as

$$\mathbf{0} \in \alpha \partial f(z) + (z - v), \tag{2.10}$$

where the linear term can be expressed as a quadratic term taking advantage of the differential operator with respect to z

$$\mathbf{0} \in \partial_z \left(\alpha f(z) + \frac{1}{2} \|z - v\|_2^2 \right). \tag{2.11}$$

Since the quadratic term is strongly convex, finding the zero of expression (2.11) necessarily yields one unique result when $\alpha > 0$. Therefore z is the result of a minimization problem which can be formulated as

$$z = \underset{x}{\operatorname{argmin}} \left(f(x) + \frac{1}{2\alpha} \|x - v\|_2^2 \right). \tag{2.12}$$

Statement (2.12) is nothing but the definition of the proximal operator with parameter α. Therefore the resolvent of the subdifferential operator of closed proper convex function is a proximal operator.

2.3 Proximal gradient method

Similarly the GFBS algorithm, the proximal gradient method is a forward-backward method. It is interesting to understand its working principle since the GFBS algorithm reduces to the proximal gradient method for a particular case. The method can be illustrated as a fixed-point method, which justifies namely what convergence criterion can be applied for such algorithms. The illustration is taken from N. Parikh and S. Boyd [7].

2.3.1 Definition

The proximal gradient method is intended to solve the following class of problems

$$\underset{x}{\text{minimize}} \quad f(x) + g(x) \ , \tag{2.13}$$

where $f : \mathbb{R}^n \to \mathbb{R}$ and $g : \mathbb{R}^n \to \mathbb{R} \cup \{+\infty\}$ are closed convex and proper. In addition, the function f is assumed differentiable. The idea behind the proximal gradient method is to split the objective

between the differentiable part f and the non necessarily differentiable part g. This allows g to be for example an indicator function of a convex set. The proximal gradient method consists of solving in an iterative fashion[7]

$$x^{k+1} := \mathbf{prox}_{\lambda^k g}\left(x^k - \lambda^k \nabla f(x^k)\right), \tag{2.14}$$

where the parameter λ^k should always be strictly positive and $k \in \mathbb{N}$ denotes the iterates.

The proximal method involves two distinct steps. A conventional gradient descent is first performed using the knowledge of ∇f. The result of the gradient step yields therefore a new point which will be closer to the minimum of f but not necessarily minimizing the cost g. For this reason, the proximal operator determines the best trade-off between staying close to the minimum of f or moving towards the minimum of g. The result of the trade-off yields therefore the new iterate and the loop is closed.

It is interesting to highlight two limiting cases. First, when the function $g = 0$, the proximal gradient method reduces to conventional gradient descent. Second, when the function $f = 0$, the proximal gradient method reduces to a proximal minimization algorithm[7]. Note also that the gradient descent step can be referred to as the forward step and the proximal step to the backward step.

2.3.2 Fixed-point method

The proximal gradient method can be interpreted as a fixed-point method to find the minimizers of (2.13). According to the definition of the fixed-point method, a point z is a solution of (2.13) if and only if the following statement is true

$$\mathbf{0} \in \nabla f(z) + \partial g(z). \tag{2.15}$$

Starting from statement (2.15), it is possible to show that the proximal gradient method is a fixed-point method. First multiplying (2.15) by a constant factor $\lambda > 0$ yields

$$\mathbf{0} \in \lambda \nabla f(z) + \lambda \partial g(z), \tag{2.16}$$

which can be reformulated as

$$\mathbf{0} \in (-z + \lambda \nabla f(z)) + (z + \lambda \partial g(z)). \tag{2.17}$$

The idea here is to make a resolvent operator appears. Therefore, statement (2.17) becomes

$$(I - \lambda \nabla f(z))(z) \in (I + \lambda \partial g)(z), \tag{2.18}$$

or more specifically

$$z = (I + \lambda \partial g)\left[(I - \lambda \nabla f(z))\,(z)\right], \tag{2.19}$$

which is the fixed-point iteration of the proximal algorithm method. Using (2.8), the following statement is obtained

$$z = \mathbf{prox}_{\lambda g}\left(z - \lambda \nabla f(z)\right). \tag{2.20}$$

The proximal gradient method is therefore a fixed-point method and the fixed-point z in (2.20) is the minimizer of problem 2.13. The fixed-point interpretation justifies meanwhile the use of an intuitive convergence criterion which would have been to check the norm of the difference between two consecutive iterates to be lower than a threshold value ϵ. Indeed, according to [9], the convergence rate of fixed-point method is at least geometric.

2.4 Generalized Forward-Backward Splitting algorithm

In [8] H. Raguet and L. Landrieu present the preconditioning of the GFBS method. Note that the present work does not aim at investigating neither the preconditioning of the method nor the conditions for convergence. However, as discussed earlier, the GFBS has interesting characteristics which motivate its implementation. The previous sections should help to understand the method. First, the algorithm is presented together with the assumptions on the problem. Second, it is shown how to apply the algorithm to the specific problem (2.2). Finally the method is compared to the ADMM algorithm in terms of privacy and variable communication.

2.4.1 Algorithm presentation

The Generalized Forward-Backward algorithm aims at solving the following class of problems

$$\mathbf{0} \in \nabla f(z) + \sum_{i=1}^{M} \partial g(z). \tag{2.21}$$

Statement (2.21) is equivalent to (2.15) when $M = 1$, where the GFBS algorithm reduces then to a proximal gradient method. Generalizing the algorithm to several agents, i.e. $M > 1$ suggests that the backward step will be performed several times according to (2.21). In order to perform M backward operations, consider $x \in \mathbb{R}^n$ as being the main variable and M auxiliary variables $(z_i)_{1 \leq i \leq M} \in \mathbb{R}^n$. For each iterate of the algorithm the backward step is performed as

$$\forall i, \quad z_i := z_i + \rho \left[\mathbf{prox}_{\frac{1}{\gamma} W_i g_i}\left(2x^{\mathcal{H}_i} - z_i - \gamma \nabla f(x)^{\mathcal{H}_i}\right) - x^{\mathcal{H}_i}\right], \tag{2.22}$$

where the variable $x^{\mathcal{H}_i}$ represent the restriction of x to the subspace $H_i \subseteq \mathbb{R}^n$ associated to the $\mathrm{i_{th}}$ agent. The auxiliary variables z_i are then gathered to form the next x iterate

$$x := \sum_{i=1}^{M} W_i z_i, \tag{2.23}$$

where W_i is a weight matrix that is a adjusted to tune the influence of each operator [8]. The step size of the proximal operator γ is kept constant along the iterates. The parameter ρ is a relaxation parameter. It is interesting to note that when $\rho = 1$, expression (2.22) becomes an ordinary proximal minimization step.

The hypothesis on the algorithm are the following:

(h1) The function $f : \mathbb{R}^n \to \mathbb{R}$ is convex and differentiable and ∇f is Lipschitz L.

(h2) For each $i \in \{1, \ldots, M\}$, $g_i : \mathbb{R}^n \to \mathbb{R} \cup \{+\infty\}$ are closed convex and proper.

(h3) There exists at least one solution to the problem such that **argmin** $\{f + \sum_i g_i\} \neq \emptyset$

The requirements on the parameters are:

(r1) The step size γ satisfies $L\gamma < 2$.

(r2) The relaxation parameter satisfies $\rho \in \left[0, 2 - \frac{1}{2}L\gamma\right]$.

(r3) For each $i \in \{1, \ldots, M\}$, $W_i \in \mathcal{S}_+^n$ such that $\sum_{i=1}^{M} W_i = I^n$.

The algorithm of the GFBS method is presented below.

Algorithm 2.1 Generalized Forward Backward Splitting.

given

$$(z_i)_{1\leq i\leq M} \in \mathbb{R}^n,\ (W_i)_{1\leq i\leq M} \in \mathcal{S}_+^n,\ \sum_{i=1}^{M} W_i = I^n$$

$$L\gamma < 2,\ \rho_k \in \left[0, 2 - \frac{1}{2}L\gamma\right],\ k \in \mathbb{N}$$

initialize

$$x := \sum_{i=1}^{M} W_i z_i,\ k = 0$$

repeat

1. **Forward step**

   $p := 2x - \gamma\, \nabla f(x)$

2. **Backward step**

   for $i = 1 \ldots M$ **do**

   $z_i := z_i + \rho_k \left[\mathbf{prox}_{\frac{1}{\gamma} W_i g_i}(p^{\mathcal{H}_i} - z_i) - x^{\mathcal{H}_i}\right]$

3. **Gather operation**

$$x := \sum_{i=1}^{M} W_i z_i$$

4. Update

$$k := k + 1$$

until convergence

return x

As discussed in the fixed-point interpretation of the proximal gradient method, the convergence criterion amounts to checking the norm of the difference between two consecutive iterates x^k and x^{k-1}. When it is less than a defined threshold ϵ such that

$$\|x^k - x^{k+1}\| \leq \epsilon, \tag{2.24}$$

it is considered that the method has converged to a point reasonably close to the real minimizer. Then the main loop is stopped and the last x^k iterate is the solution to the problem (2.21).

2.4.2 Specific problem

In order to solve (2.2) with the GFBS algorithm, its cost function must satisfy the two first cited hypothesis. For the convenience, consider again (2.2)

$$\begin{aligned} \underset{y,u,x}{\text{minimize}} \quad & \overbrace{\frac{1}{2}\left(h^\top y - r\right)^\top H \left(h^\top y - r\right)}^{f} + \sum_{i=1}^{M} \overbrace{q^\top y_i}^{g_i} \\ \text{subject to} \quad & x_i(t+1) = A_i x_i(t) + B_i u_i(t) \\ & y_i(t) = C_i x_i(t) + D_i u_i(t) \\ & x_i(t) \in X_i \\ & u_i(t) \in U_i \\ & i = 1, \cdots, M \\ & t = 1, \cdots, T \end{aligned} \quad . \tag{2.25}$$

First, f is a quadratic function, which is convex and differentiable. Its gradient is given by

$$\nabla f = hH\left(h^\top y - r\right), \tag{2.26}$$

and its Lipschitz constant L can be computed as the maximum eigenvalue of $\nabla^2 f$.

Second, the cost g_i for each agent is a linear function, which is convex by definition. For problem (2.2) it is assumed that for each agent g_i is closed and proper. In addition, each agent's subspace $\mathcal{H}_i$ satisfies

$$\mathcal{H}_i = \{z = (z_j)_{1 \leq j \leq M} \in \mathbb{R}^{TM} | z_{i \neq j} = 0\} \tag{2.27}$$

For this particular reason, the matrix W_i becomes a redundant parameter and is omitted to solve the present problem. Therefore the implemented algorithm to solve 2.25 is given below.

Algorithm 2.2 Generalized Forward Backward Splitting.

```
given
    (z_i)_{1≤i≤M} ∈ ℝ^{TM}
    Lγ < 2, ρ_k ∈ [0, 2 − (1/2)Lγ], k ∈ ℕ
initialize
        L = max{eig(∇²f)}
    x := Σ_{i=1}^{M} z_i, k = 0
repeat
 1. Forward step
      p := 2x − γ ∇f(x)
 2. Backward step
      for i = 1...M do
        z_i := z_i + ρ_k [prox_{γg_i}(p^{H_i} − z_i) − x^{H_i}]
 3. Gather operation
      x := Σ_{i=1}^{M} z_i
 4. Update
      k := k + 1
until convergence
return x
```

When the central operator has performed the forward step, it broadcasts the variable $p^{\mathcal{H}_i}$ to each local agent to perform the backward steps. Solving the proximal operator for each agent is equivalent to solve the following optimization problem

$$\begin{array}{ll} \underset{y,u,x}{\text{minimize}} & q^\top y_i + \dfrac{1}{2\gamma}\|y_i - p^{\mathcal{H}_i} + z_i\|_2^2 \\ \\ \text{subject to} & x_i(t+1) = A_i x_i(t) + B_i u_i(t) \\ & y_i(t) = C_i x_i(t) + D_i u_i(t) \\ & x_i(t) \in X_i \\ & u_i(t) \in U_i \\ & i = 1, \cdots, M \\ & t = 1, \cdots, T \end{array} . \tag{2.28}$$

The agents communicate then the result of (2.28) to the operator and the loop is closed. In other words, the only variable communicated from the operator to the agent is the result from the forward step while the only variable communicated from the agents to the operator are the results of their local proximal operators. The proximal operator plays here the role of encapsulating each local agent hiding any private information from the main operator. The constraint on the agent's privacy is then fully satisfied.

2.4.3 Alternating Direction Method of Multipliers

The GBFS algorithm is only one among a plethora of splitting algorithms that appear in the literature. The ADMM algorithm is also a splitting method that could have been used to solve the same problem. Indeed, ADMM is intended to solve the following class of optimization problems

$$\begin{array}{ll} \underset{x}{\text{minimize}} & f(x) + g(z) \\ \text{subject to} & Ax + Bz = c \end{array} . \tag{2.29}$$

This problems can be adapted to the class of problems investigated in the present work when $A = 1$, $B = -1$ and $c = 0$ and the function $g(z)$ should be replaced by a summation of functions such that $g(z) = \sum g_i(z_i)$. The ADMM method solves[2] problem(2.29) as follows

$$\begin{array}{rcl} x^{k+1} & := & \underset{x}{\text{argmin}} \left(f(x) + \frac{\rho}{2} \|x - z^k + \frac{1}{\rho}\lambda^k \right), \\ z_i^{k+1} & := & \underset{z_i}{\text{argmin}} \left(f(x) + \frac{\rho}{2} \|x^k - z_i + \frac{1}{\rho}\lambda_i^k \right), \\ \lambda^{k+1} & := & \lambda^k + \rho \left(x^{k+1} - z^{k+1} \right). \end{array} . \tag{2.30}$$

The algorithm consists of three steps: 1 x-minimization step, M z-minimization steps and the last step which is called the dual update. Note that the x-minimization step and the y-minimization steps are proximal steps according to the definition of the proximal operator.

Similarly to the GFBS algorithm, the privacy of the local agents is preserved while performing each y-minimization steps thanks to the proximal operator. The two algorithms differ in terms of communicated variables. For the ADMM algorithm, the dual variables λ needs to be communicated in addition to the x variables. This is a disadvantage for large-scale distributed applications.

Finally, note also that the x-minimization step is carried-out with a proximal operator, which is slightly more expensive to compute than the forward step in the GFBS algorithm.

Chapter 3

Micro-grid: case study

In the present chapter, the characteristics of micro-grids are first briefly introduced. The case study is presented together with the components of the micro-grid that are considered for the simulation. These components are namely loads, energy generation and storage systems. Therefore, the second part of the chapter presents how the components are modelled. The characteristics of each of them are also given, which contain important information to formulate the local optimization problems namely. Finally, the micro-grid is scaled, i.e. based on the number of loads and energy generation devices, the number of energy storage systems is defined to achieve a realistic simulation. Note that the scaling does not take into account any economical factor.

3.1 Micro-grid

Micro-grids are defined as localized and small-scale electricity grids in opposition to the large-scale centralized grid, also called the macro-grid. While the conventional centralized power stations such as coal fired, gas and nuclear power plants are classical means to generate energy at the macro-grid scale, the distributed and localized aspect that characterizes micro-grids aims at achieving several goals:

- Energy reliability and safety,
- Minimisation of transport losses,
- Cost reduction of the energy bill,
- Diversification of energy resources,
- Carbon-emission reduction.

Micro-grids interconnect loads, i.e building and offices for instance, energy generation devices and energy storage systems. Since the energy generation is often done from renewable intermittent energy resources such as wind or solar power, energy storage systems are important components of micro-grids in order to ensure its reliability. Micro-grids are typically low-voltage AC grids.

From the centralized grid's point of view, the micro-grids are seen as single entities connected by one single switch. They can therefore be disconnected from the centralized grid and operate in an autonomous way. They are then called islanded micro-grids. They can also be used to improve the grid's robustness and to dampen disturbances. This becomes particularly true with the widespread use of renewable energy sources.

3.1.1 Energy generation

The generation of energy at the micro-grid scale targets distributed energy resources in the range of 1 kW to 10000 kW[4]. Typical technologies that are used are co-generation, solar power, wind power, etc. In the present project one focuses on solar and wind power.

The power coming from the sun can be converted into electricity using the photovoltaic technology. Photovoltaic (PV) panels are distributed energy devices that can be used at the residential level, on commercial rooftops or for industrial applications as photovoltaic power stations. One advantage is that they present no fuel costs and they require small maintenance costs. Nevertheless, the energy production is intermittent with a peak around noon.

Wind turbines are, similarly the photovoltaic panels, distributed energy generation devices and they present the same advantages and disadvantages. However, unlike PV panels where peak of production are easily predictable, the peak productions of wind farms cannot be easily forecast.

3.1.2 Energy storage

While electricity is a very efficient way to transport energy, it is a very inefficient way to store it. This is why electrical energy needs to be converted into another energy source. One focuses in the present project on batteries and flywheels which are respectively electrochemical and electromechanical energy storage systems.

Electrochemical energy devices as batteries use internal chemical components to convert electrical energy into electrochemical energy. Because they have already proven their reliability and due to their low maintenance costs, batteries are widely used. Another advantage of batteries is that they can store a large amount of energy. They present however several drawbacks when connected to a micro-grid. Batteries have a short life-time, the number of charge-discharge cycles is limited and they have a slow time response for charging and discharging.

Flywheels are energy storage systems that allow the storage of electrical energy into mechanical kinetic energy. The working principle of flywheels is simple. They are made of a rotor connected to an electric DC motor or any other induction machine. The amount of energy that can be stored in the rotor is proportional to the square of the rotor's angular velocity and proportional to its inertia. They outperforms batteries in time-response but are not able to store large quantities of energy. Flywheels have several other advantages that makes them attractive candidate for micro-grid utilities. They are low-cost, robust, environmentally friendly and they can experience a very high number of charging/discharging cycles unlike batteries.

3.1.3 Case study

In the present project the Generalized Forward-Backward Splitting algorithm is used to optimally distribute a power demand profile to the heterogeneous components of a small micro-grid. The micro-grid is composed of several building offices in the neighbourhood of Lausanne, aiming at becoming more autonomous and producing part of their energy locally from renewable energy resources. The building offices are smart-buildings that can actively take part to the micro-grid regulation by adapting their heating electricity consumption but they cannot produce energy. They have to satisfy locally their constraints (e.g. room temperature) and they want this information to be kept private from the central micro-grid operator. The GFBS solver is therefore perfectly adapted to this case study.

Energy is locally produced from photovoltaic panels. Interconnected to building offices, their use makes sense because they produce energy during office hours. The energy storage systems composed of aggregated flywheels and batteries can then satisfy the lower energy demand during the night. The interaction of batteries and flywheels is interesting due to their different time-response. In practice, the flywheels can be used to optimize the charge and discharge of the batteries in order to extend their life-time and damp fast dynamic perturbations on the micro-grid.

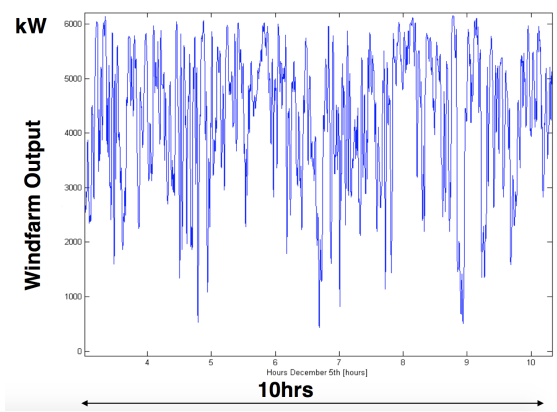


Figure 3.1: Wind-farm power output[1]

For the case study, the micro-grid is connected to the central grid for frequency stabilisation purpose. It is assumed that the macro-grid is affected by perturbations, which tends to destabilize its frequency. These perturbations can for instance arise from wind farms as illustrated by figure 3.1. To stabilize the frequency at the macro-grid level, the micro-grid operator has to track a highly varying power profile provided by the macro-grid operator. The strong assumption in the present case study is that the macro-grid operator knows perfectly the perturbations he is about to undergo in the next 24 hours, which is most probably not possible to forecast in practice.

3.2 Buildings

The buildings considered in the micro-grid are small unique story building offices of 511 m^2. They are rectangle-shaped buildings characterized by an aspect ratio of 1.5. The building model con-

siders the electricity consumption due to the heating and cooling devices as well as the electricity consumption of the building due any other devices.

3.2.1 Discrete-time model

Because the buildings play an active role in the optimization load-sharing problem, a discrete time-model is required. The small office model is provided by `OpenBuild`[6]. The model is formulated as

$$\begin{aligned} x(t+1) &= Ax(t) + B_u u(t) + B_v(t) \\ \theta(t) &= Cx(t) \end{aligned} . \tag{3.1}$$

The state variable x of the model has undefined units due to the model reduction[6]. The disturbance v is characterized by the outside temperature, the solar irradiance and the activity of the people during the day within the building. The variable of the model θ corresponds to the temperature in degree Celsius of the building's rooms. Each building has five rooms. The input u corresponds to the thermal heat in kW that is entering or leaving each room of the building. Depending on the cooling or heating applications the input of the model can respectively be negative or positive. In order to distinguish the heating from the cooling load, the inputs of the model are decoupled as,

$$\begin{aligned} x(t+1) &= Ax(t) + \begin{bmatrix} B_u & -B_u \end{bmatrix} \begin{bmatrix} u_h(t) \\ u_c(t) \end{bmatrix} + B_v v(t) \\ \theta(t) &= Cx(t) \end{aligned} , \tag{3.2}$$

where u_h and u_c denote respectively the heat load and the cooling load which are now both positive. Because these inputs do not correspond to the electricity that the buildings will buy from the micro-grid, the required thermal power must be converted into the effective electrical power consumption. One considers therefore that the building is equipped with heat pumps for both the cooling and heating, which are characterized by a coefficient of performance (COP). The heating coefficient of performance can be computed as

$$COP_h = \eta_h \frac{\theta_h}{\theta_h - \theta_c}, \tag{3.3}$$

where the hot temperature θ_h and the cold temperature θ_c are respectively assumed to be the building temperature at 25 °C and the mean outside temperature. When the efficiency η_h for heating is equal to one, it corresponds to an ideal heat pump. However, due to several factors such as the pump efficiency, pressure losses, heat losses, etc. it is a good assumption to consider that the efficiency is around 30 %. This yields a realistic result for a COP_h of about 3.5. The cooling coefficient of performance is computed similarly, except that it is computed with respect to the cold temperature and the efficiency coefficient is assumed to be the equal to η_h.

Finally the discrete-time building model becomes

$$
\begin{aligned}
x(t+1) &= Ax(t) + \begin{bmatrix} B_u & -B_u \end{bmatrix} \begin{bmatrix} u_h(t) \\ u_c(t) \end{bmatrix} + B_v v(t) \\
\theta(t) &= Cx(t) \\
y(t) &= \begin{bmatrix} COP_h & COP_c \end{bmatrix} \begin{bmatrix} u_h(t) \\ u_c(t) \end{bmatrix}
\end{aligned} \quad , \tag{3.4}
$$

where $y(t)$ denotes the total electricity consumption in kW for one single building at time t.

3.3 Batteries

The battery model is kept simple, assuming Lithium-ion batteries. The battery model does not take into account neither the DC/AC converters nor the battery system efficiency. Regarding the power convention, a negative power output corresponds to power generation because the battery discharges. On the contrary, positive power output corresponds to the storage mode. The internal resistance of the battery is taken into account in the model. Finally, it is considered that the output voltage across the battery remains always the same and is equal to the battery open-circuit potential.

3.3.1 Discrete-time model

The state variable of the system is assumed to be the state-of-charge which will be referred to as *soc*. The input of the system is the charging or discharging current denoted by I. According to the Coulomb law[5], the state-of-charge of the battery satisfies the following time differential equation

$$
\frac{d\,\mathrm{soc}}{dt} = -B\mathrm{soc} + \frac{1}{C} I, \tag{3.5}
$$

where B denotes any loss coefficient due to battery self-discharging and C denotes the battery capacity expressed in Ah. Following the above convention, a positive value of the current charges the battery and a negative value causes its discharge. Considering the internal resistance R_s of the battery, the output power of the battery system can be expressed as a function of the open-circuit potential of the battery V_{oc} and the input current I

$$
P = V_{oc} I - R_s I^2. \tag{3.6}
$$

The state variable x and input u of the continuous-time model can expressed with respect to normal operating quantities soc_N and I_N such that,

$$
x = \frac{\mathrm{soc}}{\mathrm{soc}_N} \ , \quad u = \frac{I}{I_N}, \tag{3.7}
$$

where x and u are dimensionless quantities.

Based on expressions (3.5) and (3.6), the simple continuous-time battery model is

$$\begin{array}{rcl} \dot{x} & = & \left(-\dfrac{B}{\mathrm{soc}_N}\right) x + \left(\dfrac{I_N}{\mathrm{soc}_N C}\right) u \\ y & = & (V_{oc} I_N)\, u - \left(R_s I_N^2\right) u^2 \end{array} . \tag{3.8}$$

Finally, (3.8) can be linearised around a stationary point (x_s, u_s) and a Zero Order Hold (ZOH) method is then used to compute the coefficients A, B, C and D of the discrete-time model

$$\begin{array}{rcl} x(t+1) & = & Ax(t) + Bu(t) \\ y(t) & = & Cx(t) + Du(t) \end{array} . \tag{3.9}$$

3.3.2 Battery operating constraints

It is assumed in the present model that the potential across the battery remains constant while it is not the case in reality. In fact, the battery potential is closely related to its state-of-charge that should be kept within certain bounds to avoid too low or too high potential. This is particularly true for Lithium-ion batteries[11]. This is the reason why the state-of charge is kept within 10 % to 90 %.

The C-rate of the battery express how fast the battery discharges. When the battery operates at a unity C-rate, the battery discharge or charges in one hour. Limiting the C-rate is in fact equivalent to limit the maximum and minimum input current. In order to operate the battery efficiently, a low C-rate is usually considered to avoid Joule losses caused by the internal resistance of the battery. It is considered that the batteries of the micro-grid will operate at a C-rate equal to 0.1, which means that the maximum current is limited to 10% of the battery capacity C. Note also that an additional constraint on input current rate of charge is added to the problem, limiting absolute changes of 5 A, which should smooth the charging and discharging of the battery.

3.4 Flywheels

The working principle of a flywheel is simple. A rotating mass called the rotor rotates very fast around its axis. Kinetic energy E_k is stored in the rotor of the flywheel. The rotor is characterized by an inertia J and its angular speed ω

$$E_k = \frac{1}{2} J \omega^2. \tag{3.10}$$

Considering that the rotor of the flywheel can be modelled by a hollow cylinder, the inertia J of the rotor can be expressed according its outer, inner radius and height respectively given by r_{out}, r_{in} and h

$$J = \frac{\pi \rho h}{2} \left(r_{out}^4 - r_{in}^4\right). \tag{3.11}$$

Therefore, low-speed flywheels will maximize the amount of energy stored in the system by increasing the inertia of the rotor. This is done by using high density materials such as steel. Fast-speed flywheels use rather a small rotor made of composite materials such as Kevlar or Carbon-Fiber[10] but they target higher angular velocities.

To model the flywheel system, it is assumed that there are no friction losses. They can be neglected for two reasons. First, for some flywheels systems, the rotor is worn by a magnetic bearing technology, which minimize the friction losses between the rotor and its support. Second, the rotor is sometimes also installed in a vacuum chamber such as to minimize the losses caused by the drag of air.

3.4.1 Discrete-time model

A simple continuous-time model for the flywheel is obtained by considering the momentum conservation on the rotor

$$J\frac{d\omega}{dt} = -B\omega + \tau + \tau_f, \tag{3.12}$$

where the B is a loss coefficient, τ_f represents the torque due any friction losses, τ is the input torque of the system and ω is the angular velocity of the rotor. For a flywheel system it is reasonable to assume that τ_f is negligible, as it has been discussed previously.

The output of the system is the power delivered or stored by the flywheel. Similarly to the battery input current, a positive or negative input torque τ reflects respectively the fact that the flywheel is storing or producing energy and can be expressed as

$$P = \omega\tau, \tag{3.13}$$

where P denotes the power in kW that is stored or provided by the flywheel. Similarly to the battery model, it is important to highlight the fact that expression (3.13) does not take into account the losses caused by power converters and other mechanical components present in the flywheel system such as gears, etc.

One defines respectively the state variable of the system x and the system input u with respect to the nominal angular velocity ω_N and the nominal torque τ_N of the system

$$x = \frac{\omega}{\omega_N},\ u = \frac{\tau}{\tau_N}, \tag{3.14}$$

which are dimensionless variables and the output of the model y is the power P in kW.

The continuous-time model becomes

$$\begin{array}{rcl} \dot{x} & = & -\left(\frac{B}{J}\right)x + \left(\frac{\tau_N}{J\omega_N}\right)u \\ y & = & (\omega_N T_N)\,ux \end{array}. \tag{3.15}$$

The continuous-time model can be linearised around a stationary point (x_s, u_s) and the discrete model can be obtained using a ZOH method to compute the matrices A, B, C *and* D of the discrete model

$$\begin{array}{rcl} x_{k+1} & = & Ax_k + Bu_k \\ y_k & = & Cx_k + Du_k \end{array} . \tag{3.16}$$

3.4.2 Flywheel operating constraints

To operate flywheels safely, the angular velocity of the rotor is upper and lower bounded. Due to centrifugal effects the rotor might dislocate for high tip rotor velocities. More specifically, the tensile strength of the rotor material limits its maximum angular velocity. There exist also a lower bound on the angular velocity[10] to limit the discharge depth of the flywheel. The total energy E_{tot} that can be stored depends therefore on the minimum and maximum angular velocity allowed for a given rotor respectively denoted by $\underline{\omega}$ and $\overline{\omega}$

$$E_{tot} = \frac{1}{2} J \left(\overline{\omega}^2 - \underline{\omega}^2 \right) . \tag{3.17}$$

Similarly to the batteries, flywheels have a C-rate that limits their rate of charge and discharge. In order to impose a C-rate, the input torque of the system is limited to a minimum and maximum value that depends on the total energy storage capacity E_{tot} and the nominal angular speed.

3.5 Photovoltaic panels

At the micro-grid scale, the photovoltaic panels play the role of passive agents. They are considered as negative loads introducing a disturbance on the grid by producing energy whenever they can. The energy produced from the PV panels can be estimated with a certain level of accuracy.

3.5.1 Photovoltaic panels modelling

The energy production from PV panels depends on the day time, the earth's position towards the sun and the location of the panels themselves on earth. The variable of interest to predict the PV panel power production is the solar irradiance at the surface of the earth. The solar irradiance can be estimated as a function of the zenith angle Z

$$I = S \cos(Z) , \tag{3.18}$$

where I and S are expressed in $\mathrm{W/m^2}$ and S is the solar constant approximately equal to 1000 $\mathrm{W/m^2}$. The value of the zenith angle Z depends on the latitude, time of the day and the solar declination angle, which is different for each day of the year. According to [3], the zenith angle can be computed as

$$Z = \cos^{-1}\left(\sin\Phi\sin\delta + \cos\Phi\cos\delta\cos H\right), \tag{3.19}$$

where Φ denotes the location's latitude, δ denotes the solar declination angle and H the hour angle. First, the declination can be computed according to the current day n_{day} of the year as [3]

$$\delta = 23.45\sin\left(2\pi\frac{284 + n_{day}}{365}\right), \tag{3.20}$$

and the hour angle is computed as

$$H = \frac{\pi}{6}\left(12 - t\right), \tag{3.21}$$

where t is the day time expressed in hours.

Using these relationships makes it possible to have an estimation of the solar irradiance at a particular location for a given day.

With an expression of the solar irradiance as a function of time, it is possible to estimate the instantaneous PV panels power production for a given place and a given panels' area. However, it must be taken into account that the estimation of the earth's surface irradiance is likely to be less due to albedo. Albedo characterizes reflection from the sun rays on the atmosphere, clouds, etc. This is why the effective irradiance is often less than the estimated one. The effect of albedo can be accounted as an efficiency factor denoted by η_a. In addition to that, the PV panel power production is limited by the efficiency η_{pv} of the system, i.e. the ability at converting sunlight into electricity. For modern systems, it is reasonable to assume an efficiency of about 20 %.

The instantaneous power generated per square meter from PV panels can be expressed as

$$P_{pv} = \eta_{pv}\eta_a I(n, t, \Phi). \tag{3.22}$$

Note that for the simulation, the albedo efficiency is modelled as the inverse of a uniform distribution defined between a lower bound of 0.4 and upper bound of 0.6, which would correspond to a slightly cloudy day for instance. Note also the the latitude of Lausanne is equal to 46 °.

3.6 Problem scaling

In the previous sections, the various components of the micro-grid were explained and modelled. It is now necessary to scale the micro-grid. The number of buildings must be given together with the number of batteries, flywheels and PV panels. The simulation day is important for the building models and for the photovoltaic model and is set as the 25th January of 2000. Assume that the location of the simulation is in the neighbourhood of Lausanne.

3.6.1 Building scaling

First, the buildings consume electricity due to lighting and plug loads with an average of 10.50 W/m^2. Figure 3.2 shows a one-day estimation of the electricity consumption for an office. Note the two maxima around 10 am and 3 pm are related the significant human activity. Outside the office hours, the electricity consumption is lower. In addition to that, electricity is consumed by the heat pumps to cool down or heat up the buildings. Figure 3.3 shows the outside temperature from experimental data integrated to the `OpenBuild` toolbox[6]. It is assumed in general that the electricity contribution for heating/cooling is about 4.0 W/m^2. It could be more during the winter and less during the summer.

The micro-grid gathers 11 offices of 28 people, each made of 511 m^2. Therefore the average total energy needed for one building is about 180 kWh each day, which makes approximately a daily energy consumption of about 1980 kWh. The eleven buildings are grouped into seven sub-groups. Namely, two groups with two offices, one group with three offices and four single offices. Finally, among the seven building offices agents, two different room temperature constraints are considered. Some of the buildings decide to impose a minimum and maximum temperature of respectively 20 and 23 °C while others impose a minimum and maximum temperature of respectively 19 and 24 °C .

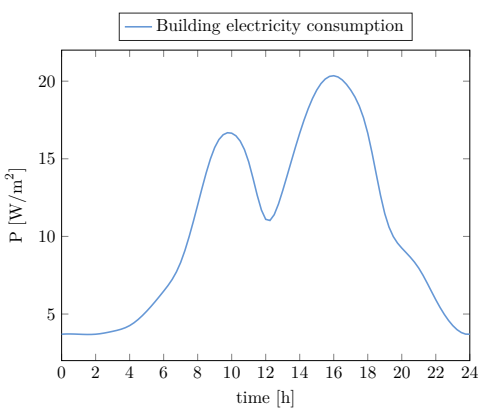


Figure 3.2: Building electricity consumption estimation in W/m^2

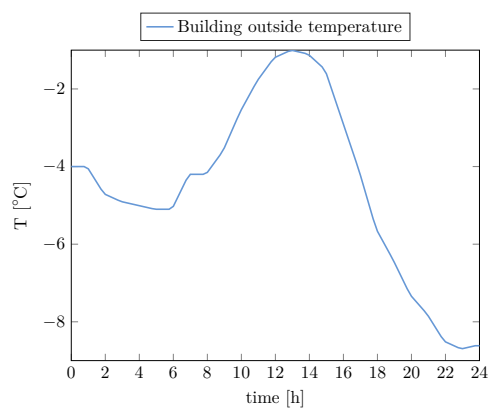


Figure 3.3: Lausanne outside temperature for the 25th January of 2000

3.6.2 PV panels scaling

To scale the total area of PV panels needed in the micro-grid, it is necessary to take into account the seasonal factor of solar power production. A good choice is to scale the total area of PV panels for the yearly mean solar irradiance. The integration and daily averaging of expression (3.18) yields 7.2 kWh/day/m^2 of sun energy that would be possible to harvest in one day under ideal conditions without albedo and without taking into account the efficiency of the PV panels. Assuming optimistically that the albedo would halve this quantity on a year average basis and considering the PV efficiency of 20 %, this result becomes 0.72 kWh/day/m^2. To ensure the micro-grid a certain level of sell-sufficiency, the area of required PV panels should satisfy the building

average consumption of 1980 kWh/day. To achieve this, a total PV panel area of 2750 m^2 is required.

Regarding the simulation, it is assumed that the micro-grid operator can predict perfectly the irradiance and albedo. Figure 3.4 shows the difference between ideal conditions and more realistic conditions. It shows namely the effective irradiance that will provide the power to the PV panels for the simulation. The effect of albedo is taken into account a computed a uniform distribution with lower and upper of 0.4 and 0.6 respectively to cut the ideal irradiance by a factor of 0.5 approximately.

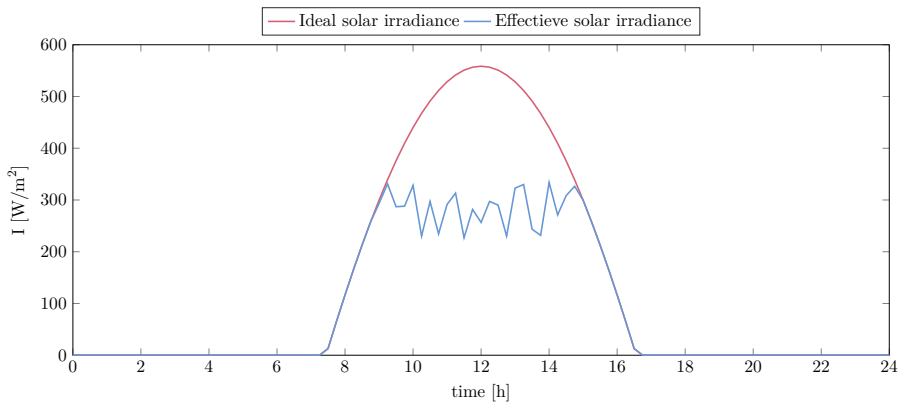


Figure 3.4: Solar irradiance Lausanne on 25th January 2000

3.6.3 Energy storage scaling

The batteries have a storage capacity of 100 kWh and the flywheels have a storage capacity of about 25 kWh. It has been decided that the energy storage systems would have together a storage capacity of 1500 kWh. Two thirds of the total available storage capacity should be ensured by the batteries and the rest by the flywheels. One agent consists therefore of a group of 20 flywheels and another agent is made of 10 batteries. At the beginning of the simulation, each flywheel and battery is assumed to dispose over half of its storage capacity.

3.6.4 Micro-grid overview

Simulation characteristics		
Data	25 january 2000	
Location	Lausanne	
Time	00:00 - 24:00	
Sampling time	15	min
Horizon	96	–
A	$\{0, 500, 1000, 1500\}$	-
Small offices		
Area per unit	511	m^2
Minimum temperature	20/19	°C
Maximum temperature	23/24	°C
Heat pump COP_{hot}	3.0	–
Heat pump COP_{cold}	3.0	–
Battery		
Number of systems	10	#
Energy storage capacity	100	kWh
Minimum state-of-charge	0.1	%
Maximum state-of-charge	0.9	%
C-rate	0.1	–
ΔI	5	A
Material	Lithium-ion	
Flywheel		
Number of systems	20	#
Energy storage capacity	25	kWh
Minimum angular speed	4000	rpm
Maximum angular speed	16000	rpm
C-rate	2	–
Rotor density	1500	kg/m^3
Rotor inner radius	25	cm
Rotor outer radius	40	cm
Rotor height	200	cm
Rotor material	Carbon fibber	
Photovoltaic panels		
Number of PV pannels	2750	#
Area per unit	1	m^2
PV efficiency	0.20	–
Estimated albedo	$\mathcal{U}(0.4, 0.6)$	-

Table 3.1: Micro-grid case study overview

Chapter 4

Micro-grid: simulation

In the previous chapters, the Generalized Forward-Backward Splitting method has been introduced and explained as well as the micro-grid case study. In the present chapter, the details for the simulation case study and algorithm are given. Then, several simulations are carried out for different power profiles. From a flat to highly varying profile, in agreement with the application of the micro-grid as frequency stabilizer.

4.1 Simulation implementation

This section aims at explaining how the simulation is implemented. First, the parameters of the reference optimization problem are given and explained in the context of the micro-grid case study as well as the parameters for the GFBS algorithm. It is then discussed how the proximal problems are solved and how different power profiles are generated.

Regarding the software support, note that the GFBS algorithm is implemented in `MATLAB` as well as the different discrete models of the micro-grid case study. The proximal optimization problems are exported from `MATLAB` to `C` using the `SPLIT` toolbox[12].

4.1.1 Case study and algorithm parameters

Consider again problem (2.2)

$$
\begin{aligned}
\underset{y,u,x}{\text{minimize}} \quad & \frac{1}{2}\left(h^\top y - r\right)^\top H \left(h^\top y - r\right) + \sum_{i=1}^{M} q^\top y_i \\
\text{subject to} \quad & x_i(t+1) = A_i x_i(t) + B_i u_i(t) \\
& y_i(t) = C_i x_i(t) + D_i u_i(t) \\
& x_i(t) \in X_i \\
& u_i(t) \in U_i \\
& i = 1, \cdots, M \\
& t = 1, \cdots, T
\end{aligned}
\tag{4.1}
$$

where the variable $y_i(t) \in \mathbb{R}$ represents the power consumption/generation in kW of the $\mathrm{i_{th}}$ agent at a given time t. The vector $y \in \mathbb{R}^{TM}$ gathers the power consumption of all the agents over the simulation time horizon. Note that the variable y is the public variable which is communicated and for each agent the state x and input u are private variables as well as the data of the discrete-time models.

Referring back to the definition of the case study in the previous chapter and in particular to table 3.1, the length of the time horizon T is equal to 96. The number of agents M is equal to 9. The batteries and flywheels are aggregated in two separate agents. The buildings form together seven agents, i.e. three single offices, two groups and two offices and one group of three offices. The collaboration parameter σ is set to 20 for the simulation. The vector $q \in \mathbb{R}^T$ is kept constant over the horizon with a unity value. Finally, the target parameter $r \in \mathbb{R}^T$ represents the power profile in kW for each time t of the day.

Regarding the parameters of the GFBS algorithm, γ and ρ must be defined according to the algorithm requirements. The parameter γ must satisfy $\gamma \leq 2/L$. For the simulation it is considered that $\gamma = 1Ł$. Therefore, the requirement on the over-relaxation parameter $\rho \in \left]0, 2 - \frac{1}{2}\right[$ simplifies to $\rho \in \,]0, 2[$ and is chosen to be equal to 1.5 for the simulation.

4.1.2 Proximal steps

For the simulation, the agents are build from a multitude of batteries, flywheels or buildings. Therefore, the proximal problem (2.28) given in the second chapter requires some modifications. Consider a discrete-time model with state matrix A, input matrix B, throughput matrix C and output matrix D. The aggregation of N models is obtained as

$$
\begin{aligned}
x(t+1) &= \mathcal{A}x(t) + \mathcal{B}u(t) \\
y(t) &= \mathcal{C}x(t) + \mathcal{D}u(t)
\end{aligned}
\quad ,
\tag{4.2}
$$

where

$$
\mathcal{A} = \begin{bmatrix} A_1 & & \\ & \ddots & \\ & & A_N \end{bmatrix}, \ \mathcal{B} = \begin{bmatrix} B_1 & & \\ & \ddots & \\ & & B_N \end{bmatrix}, \ \mathcal{C} = \begin{bmatrix} C_1 & \cdots & C_N \end{bmatrix}, \ \mathcal{D} = \begin{bmatrix} D_1 & \cdots & D_N \end{bmatrix}. \tag{4.3}
$$

The proximal operator solves therefore the following optimization problem for each aggregation of flywheels and batteries.

$$\begin{array}{ll} \underset{y,u,x}{\text{minimize}} & q^\top y_i + \frac{1}{2\gamma}\|y_i - p^{\mathcal{H}_i} + z_i\|_2^2 \\ \\ \text{subject to} & x_i(t+1) = \mathcal{A}_i x_i(t) + \mathcal{B}_i u_i(t) \\ & y_i(t) = \mathcal{C}_i x_i(t) + \mathcal{D}_i u_i(t) \\ & x_i(t) \in \mathcal{X}_i \\ & u_i(t) \in \mathcal{U}_i \\ & i = 1, \cdots, M \\ & t = 1, \cdots, T \end{array} , \tag{4.4}$$

where $\mathcal{X}$ and $\mathcal{U}$ corresponds to the aggregated constraints. The constraints on state variable x for the batteries and flywheels correspond respectively on the lower and upper tolerated value for the state-of-charge and angular velocity. Regarding the input current of the batteries and the torque of the flywheels they are also lower and upper constrained and the battery input current as an additional rate-of-change constraint. Regarding the buildings, the proximal problem to solve is almost identical to (4.4) except the fact that the discrete-time model has an additional disturbance matrix B_v and that the temperature θ of the rooms is constrained rather than their state x.

The `SPLIT` toolbox is used to co-generate every proximal problem of the case study before the simulation is run. During the simulation, for each iterate the operator communicates only parameter p to the agents, which solve problem (4.4) with the ADMM algorithm and return then its optimal power consumption y_i. In order to improve the time efficiency, each local proximal step is warm-started with the previous iterate solution.

4.1.3 Power profile

The micro-grid is connected to the centralized grid for frequency stabilization. As discussed in the previous chapter, one assumes that the macro-grid operator knows perfectly the disturbance he is going to undergo in the next 24 hours. He provides therefore the micro-grid a power profile to track such as to dampen the disturbance on the macro-grid. Note that the question here is not about the credibility of such a disturbance forecast but rather about how the ability of the algorithm to track a power profile with increasing difficulty.

The modelling of the power profile r to track is done in two step. First, the base-line power consumption/generation is computed based on the expected power that the micro-grid needs to buy or sell for the simulation. Second the perturbation is added to the base-line and is modelled as a uniformly distributed signal with zero mean. The signal $r(t)$ becomes

$$r(t) = B + A_j s(t), \tag{4.5}$$

where B is a scalar value, A_j a scaling parameter and j the index of the simulation. The perturbation s is modelled as,

$$s(t) = \mathcal{U}(t, -1, 1), \tag{4.6}$$

where it is assumed that s satisfies $0 = \frac{1}{T}\sum_{t=1}^{N} s(t)t_s$ with t_s the sampling time.

To estimate the amount of energy required to cover the needs for 25th of January 2000, an energy balance is computed in (4.7). The balance indicates the micro-grid operator should buy at least 80 kWh from the grid for that day. Over the entire day, this corresponds to a mean power of 3.33 kW. For the simulation $B = 30$ kW is chosen such that the energy-storage systems are not completely empty or over-loaded at the end of the simulation. Note that the energy storage systems start the simulation with half of their storage capacity.

Buildings	−1980	kWh
Solar	+1150	kWh
Flywheels	+250	kWh
Batteries	+500	kWh
Balance	−80	kWh

(4.7)

4.2 Simulation results

This sections presents the simulation results for different power profiles. At the micro-grid level, the quality of the power profile tracking is evaluated and the contribution for each type of agent is discussed. Then, at the agents' level the behaviour of a flywheel, a battery and a building is discussed for one simulation.

4.2.1 Power profile

Four simulations are performed, with profiles A_j, $j = 1, \cdots, 4$, which are respectively equal $0, 500, 1000$ or 1500. The amplitude of perturbation correspond to realistic grid perturbations[1].

For a flat power profile tracking, it is expected that the GFBS method will not encounter any difficulty because all the agents of the micro-grid should satisfy their constraint easily. For the second simulation the maximum amplitude of the perturbations is about $1000kW$, the micro-grid should be able to track the power profile thanks to the dynamic of the flywheels. Indeed, with a C-rate equal to 2, the maximum dynamic they can track is exactly $1000kW$. Coming to the third and fourth simulation, the maximum amplitude of the perturbations are about $2000kW$ and $3000kW$ respectively. It is there expected that the agents of the micro-grid will activate their constraints.

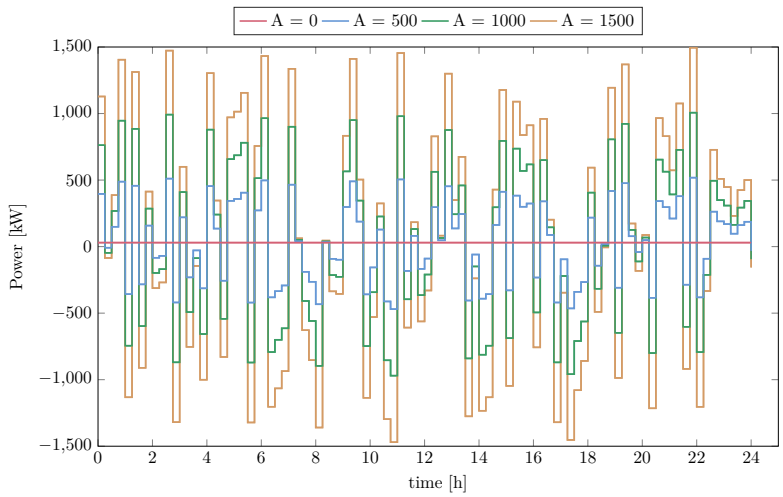


Figure 4.1: Different tracking signals as a function of parameter A_j

4.2.2 Tracking the target

Figure 4.2 shows that the flat power profile is well tracked. It will be interesting to see below the repartition of each agent's contribution on the micro-grid for the flat profile where the only perturbation arises from the power generation of the PV panels. For the second power profile, figure 4.3 shows that the algorithm achieves a good quality tracking because there is no clear distinction between the target and the contribution of the agents.

The quality of the tracking is lower when the profile varies more significantly. Figure 4.4 shows an overall good tracking behaviour but there are several weak points. Namely around 7 am and around 6 pm where there is a clear distinction between the target and the micro-grid power. Figure 4.5 shows a very low quality tracking where the micro-grid cannot cope with the too high amplitudes of the target.

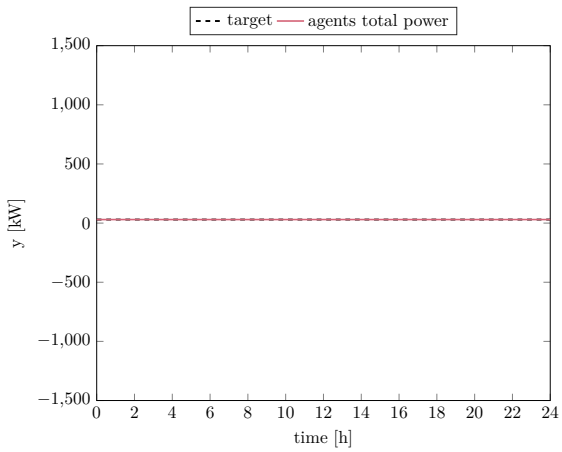


Figure 4.2: Simulation 1, target r vs. total aggregated power of the agents

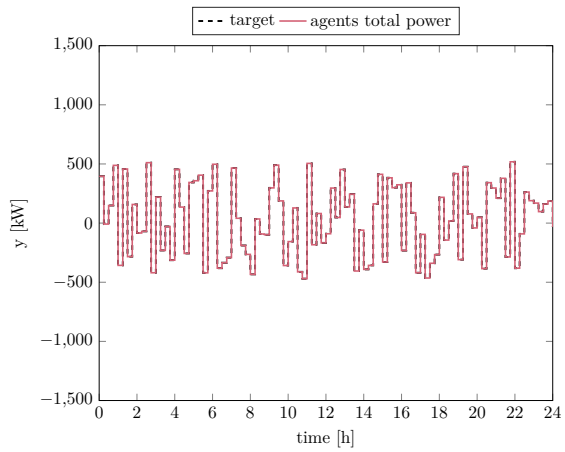


Figure 4.3: Simulation 2, target r vs. total aggregated power of the agents

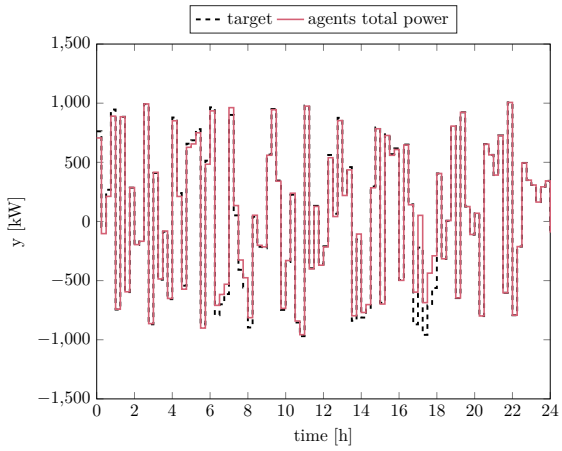


Figure 4.4: Simulation 3, target r vs. total aggregated power of the agents

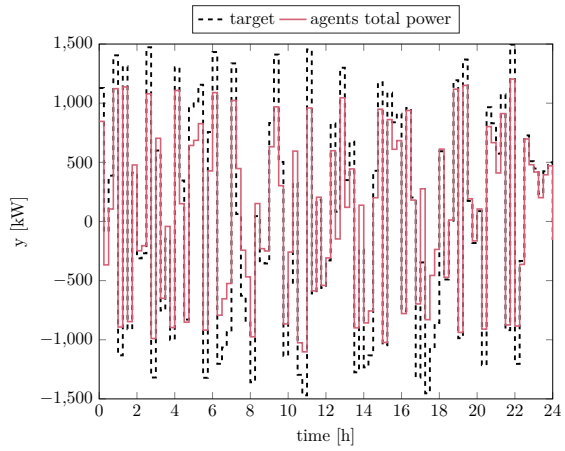


Figure 4.5: Simulation 4, target r vs. total aggregated power of the agents

4.2.3 Load sharing

It is now interesting to see how the GFBS algorithm manages the load sharing in the micro-grid. Starting first with the flat power profile, figure 4.6 shows the energy production from the PV panels. This is namely the sole disturbance that the micro-grid has to cope with. It is very clear that the building will mainly catch this internal perturbation while the batteries and flywheels will rather charge during this period of time to release energy during the night. Note that the battery contribution cannot be seen on the graphs because it exactly corresponds to contribution of the flywheels. It seems that the load is therefore equally shared among similar agents, as long as constraint are not activate.

Comments for figure 4.7, 4.8 and 4.9 are the same. First, notice that the flywheel dynamic are much faster than the other agents'. As expected, they play a major role for power profile tracking. The batteries on the contrary have slower dynamics, i.e. they have slower charge and discharge cycles

than the flywheels. Finally, one can see that the buildings play a small but non negligible role in the tracking. Consider figure 4.7, when the target is highly positive, the overall power consumptions of the buildings clearly increases.

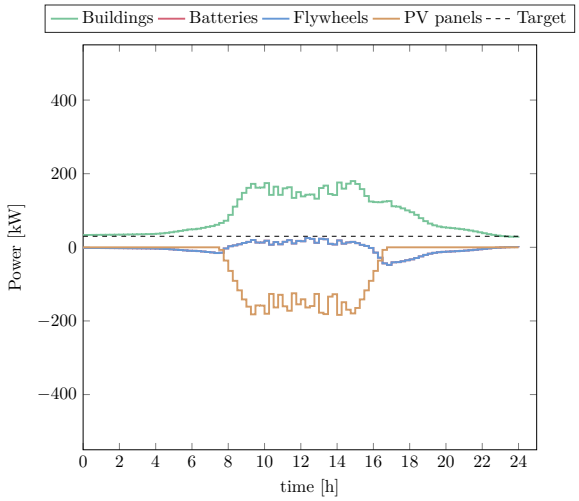


Figure 4.6: Simulation 1, power profile per agent type

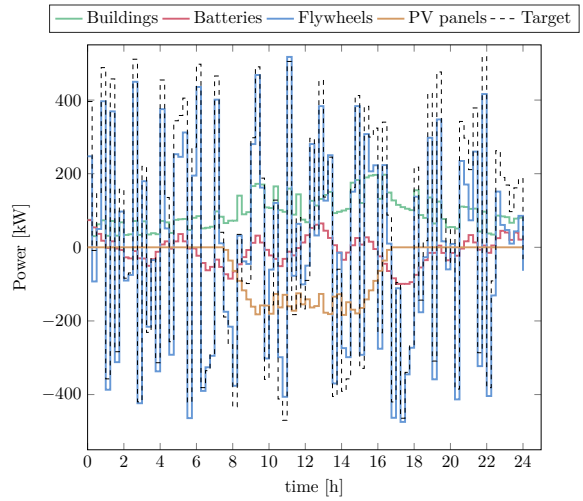


Figure 4.7: Simulation 2, power profile per agent type

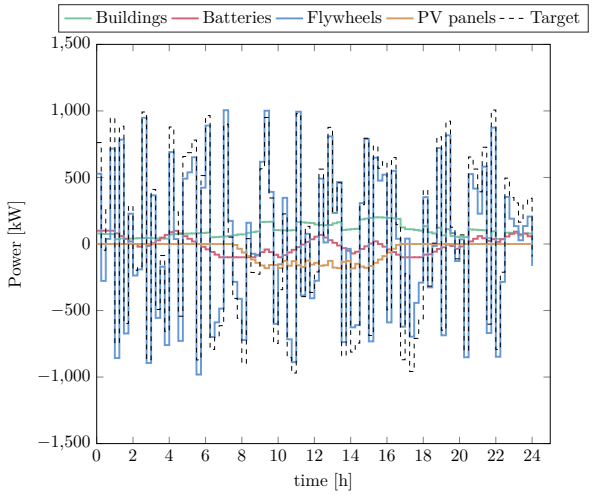


Figure 4.8: Simulation 3, power profile per agent type

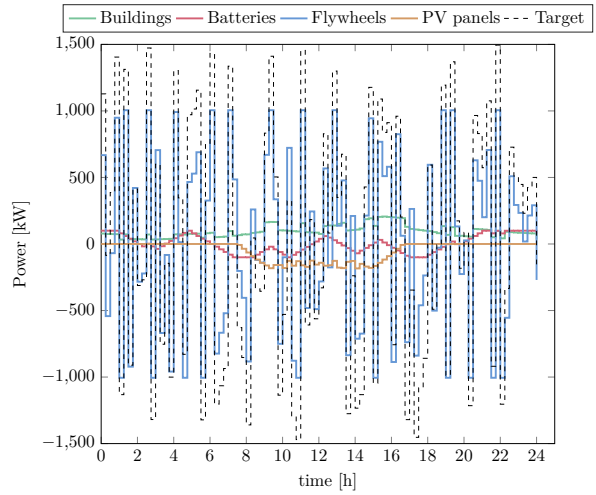


Figure 4.9: Simulation 4, power profile per agent type

4.2.4 Energy storage systems

The behaviour of the energy storage devices will be compared for simulation 3. Figure 4.10 shows the battery state-of-charge. As expected, the charging/discharging rate is small. It shows namely that for one day, the battery performs almost one discharging cycle. On the contrary, it can

be seen in figure 4.11 that the flywheels have much faster dynamics and will undergo several charging/discharging cycles over the same day. Regarding the battery constraint, it can be seen the the lower bound on the state-of-charge is activated only once at around 6 pm while the constraints on the angular velocity of the flywheels are activated several times.

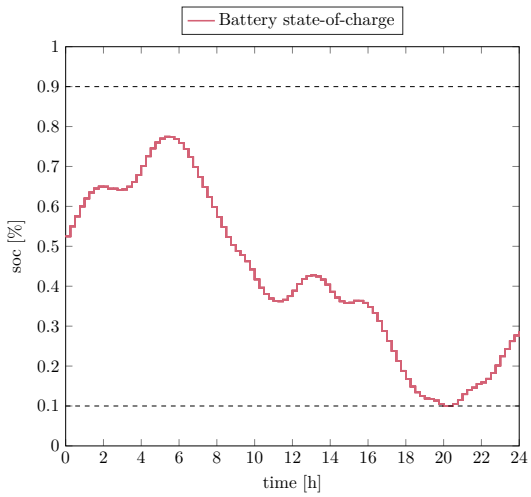


Figure 4.10: Simulation 3, battery state-of-charge

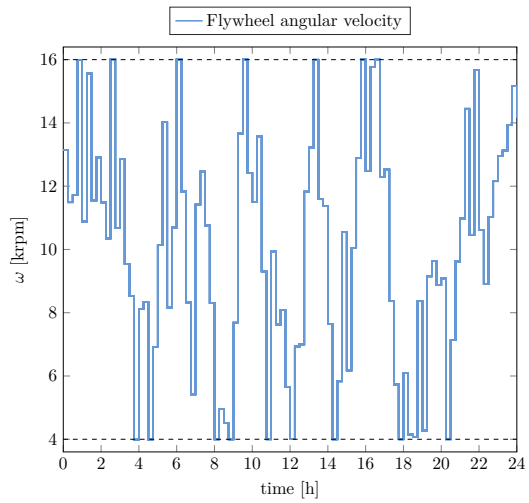


Figure 4.11: Simulation 3, flywheel angular velocity

Regarding now the input current and input torque of the batteries and flywheels, they are respectively shown in figures 4.12 and 4.13. First, the influence of the constrained rate-of-change on the input current of the batteries is clearly visible, about 5 A as expected. The lower bound of the input current is activated three times, around 8 am, 10 am and around 6 pm. Concerning the flywheels, the input torque constraints are activated several times.

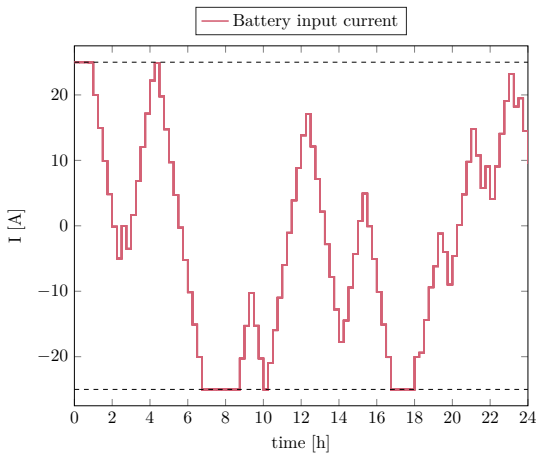


Figure 4.12: Simulation 3, battery input current

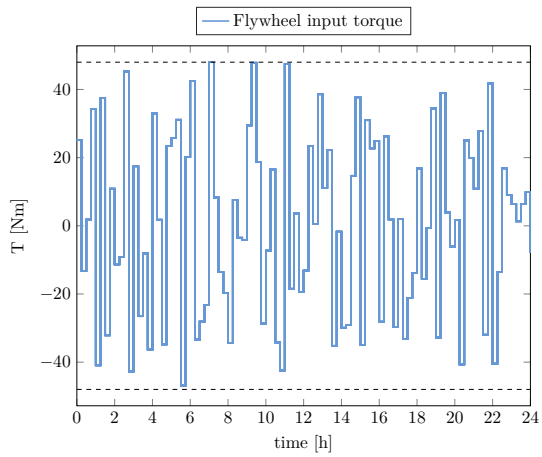


Figure 4.13: Simulation 3, flywheel input torque

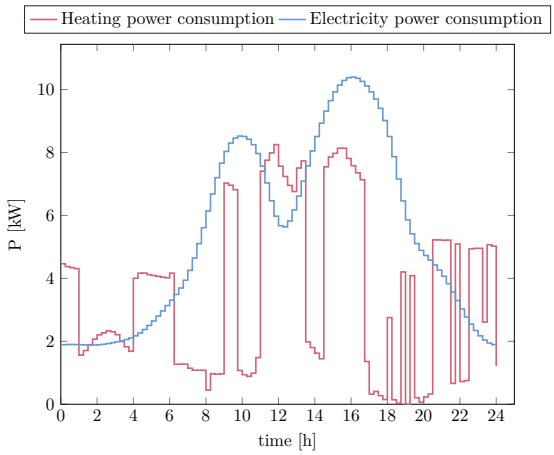


Figure 4.14: Simulation 3, office power consumption

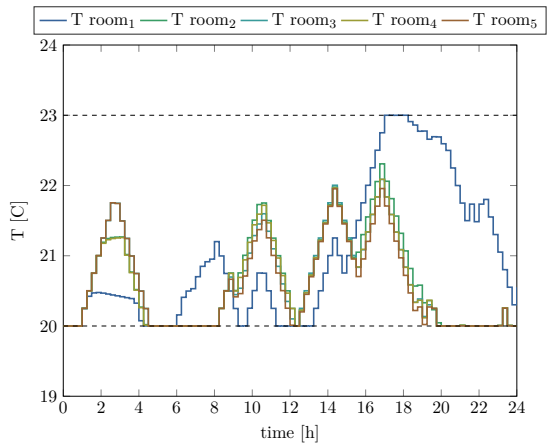


Figure 4.15: Simulation 3, office room temperature

4.2.5 Buildings

Figure 4.14 shows the power consumption of a small office. The distinction is done in the graph between the electrical consumption due to lighting or plug-in loads and the power consumption due to heating. The figure illustrates clearly that the offices contribute to the power-profile tracking. High peaks of several kWs are observed. Figure 4.15 shows the evolution of the power profile across time. It can be seen that the local optimizer tries to minimize the power consumption whenever possible.

4.2.6 Convergence

Finally, figure 4.16 shows the evolution of the residuals $\|x^k - x^{k-1}\|$ along the iterations. It can be seen that the overall trend is positive as they all seem to converge at or below $1 \cdot 10^3$ for 50

iterations. When the power profile remains constant, it can be seen that the convergence rate is much faster. On the contrary, when the power profile varies over the time horizon, the residuals seem to converge slowly. This is particularly the case for simulation 3.

The evolution of the cost function along the iterations is shown in figure 4.17. It can be seen that that they stabilize after some iterations around an equilibrium value. Notice that the cost for A_1 and A_2 are the same. This could be related to the fact that the power-profile are well tracked and consequently the quadratic term of the cost function cancels for both cases. On the contrary, for the two last simulations, the cost function is higher, reflecting therefore a low quality tracking. The cost for A_3 is less than the one for A_4 because the last simulation presents more severe amplitudes in the power profile.

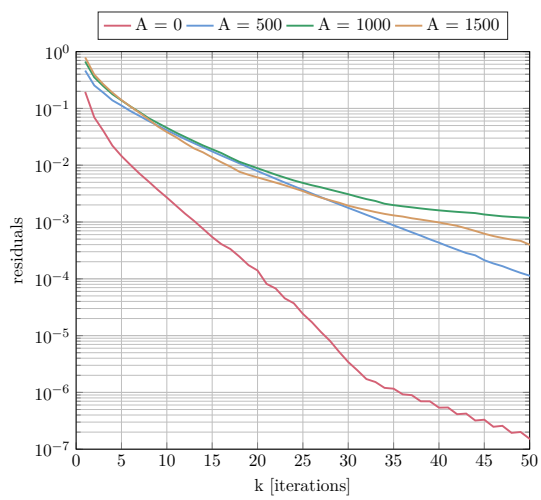


Figure 4.16: $\|x^k - x^{k-1}\|$ along the iterations

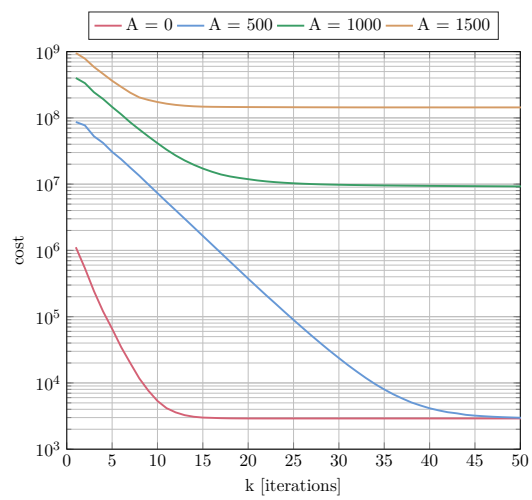


Figure 4.17: Cost function $f + \sum g_i$ along the iterations

Chapter 5

Conclusion

To conclude, in the present project a Generalized Forward-Backward Splitting (GFBS) method is studied, interpreted and implemented. The splitting method is suited for distributed convex optimization problems composed of one differentiable part and a sum of possibly non-smooth functions. To test the ability of the method to solve large-scale distributed optimization problems, a micro-grid simulation is built interconnecting building loads and various energy production and storage devices. The optimization problem is formulated such that the components of the micro-grid have to collaborate in order to track a given power profile.

The Generalized Forward-Backward Splitting method can be characterized as a proximal algorithm. When the problem is composed of only two terms in the objective, the method can be seen as a proximal gradient method where the forward step corresponds to the gradient descent and the backward step corresponds to the evaluation of the proximal operator. The backward step is parallelizable and can be distributed over all the agents of the problem. Its fixed-point interpretation gives a clear understanding of what stopping criterion can be used. Regarding the agents' privacy, it is shown how the proximal steps protect the private variables of the agents. Lastly the GFBS algorithm requires less variable communication than other splitting methods such the Alternating Direction Method of Multipliers, which involves the additional communication of the dual variables.

The micro-grid case study interconnects building offices, batteries, flywheels and PV panels. The buildings actively take part to the micro-grid regulation by adapting their power consumption depending on the available power on the micro-grid. The interaction of flywheels and batteries is complementary and interesting for two reasons. First, the flywheels can cope with a high number of charging and discharging cycles over a short period of time while batteries cannot. Second, batteries can store more energy over a longer time period than flywheels. The PV panels are connected as passive agents to the micro-grid, i.e. they produce energy whenever they can. For the case-study, the micro-grid is connected to the central grid where it plays the role of a frequency stabilizer entity.

Finally, the GFBS method is implemented in `MATLAB` where the solving procedure for the backward steps are exported in `C` using the `SPLIT` toolbox. The simulations are performed for several power profiles, from flat to strongly varying profiles. The results for the simulations show that the more abrupt is the profile to track, the higher is the final simulation cost, the more agents' constraints are activate and subsequently the more iterations for the algorithm to converge.

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